

ARXIV 2606.24093 · COMPUTATIONAL LITERARY HISTORY

Predicting Poets' Origins from Verse

Regional linguistic fingerprints in the *Complete Tang Poems* 全唐詩

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鄉
音
無
改
鬢
毛
衰

詩

Can a poem betray where its poet came from?

少小離家老大回

鄉音無改鬢毛衰

*"I left home young and return an old man;
my native accent is unchanged, though my temple-hair has thinned."*

— He Zhizhang 賀知章

"On Returning Home" 回鄉偶書



An accent survives a lifetime away from home. Does home also survive on the page?

Literary historians have long debated regional schools and local poetic traditions in Tang China — yet whether geographic origin shows up as quantifiable patterns in poetic language has resisted systematic study.

This paper treats it as a measurement problem: **read ~49,000 poems, and try to predict each poet's home circuit from text alone.**

(Keep He Zhizhang in mind — he returns near the end.)

Three questions, asked of 49,000 poems



RQ1

Prediction

Can a model predict a poet's circuit of origin from their collected work alone?



RQ2

Signal

Which features — lexical choice, imagery, tonal patterns, thematic preference — carry the strongest regional signal?



RQ3

Persistence

Does any regional fingerprint persist across the dynasty's poetic eras?

Framed for interpretability throughout — **every modelling choice favours features a literary historian can read and contest.**

The Complete Tang Poems 全唐詩

900

volumes, aggregated end-to-end

≈49,000

poems parsed into poet-level corpora

2,200+

poets in the full anthology



Source & digitisation. Texts from the Chinese Text Project (中國哲學書電子化計劃) — the standard open digital edition of the Qing-era anthology.



Pipeline sanity check. Bai Juyi (白居易) — ≈2,600 poems recovered end-to-end, confirming the parser holds up at scale.

全
唐
詩

Grounding 357 poets in ten circuits 道



Aggregate

~49,000 poems grouped into one corpus per poet



Locate

Birthplaces linked via the China Biographical Database (CBDB)



Filter

Keep poets with ≥ 5 surviving poems and unambiguous attribution



Label

357 poets · 10 circuits — 242 in the South/North subset

The ten early-Tang administrative circuits

關內 Guannei ♦

河南 Henan ♦

河東 Hedong

河北 Hebei

隴右 Longyou · 6

山南 Shannan

淮南 Huainan

江南 Jiangnan · 126

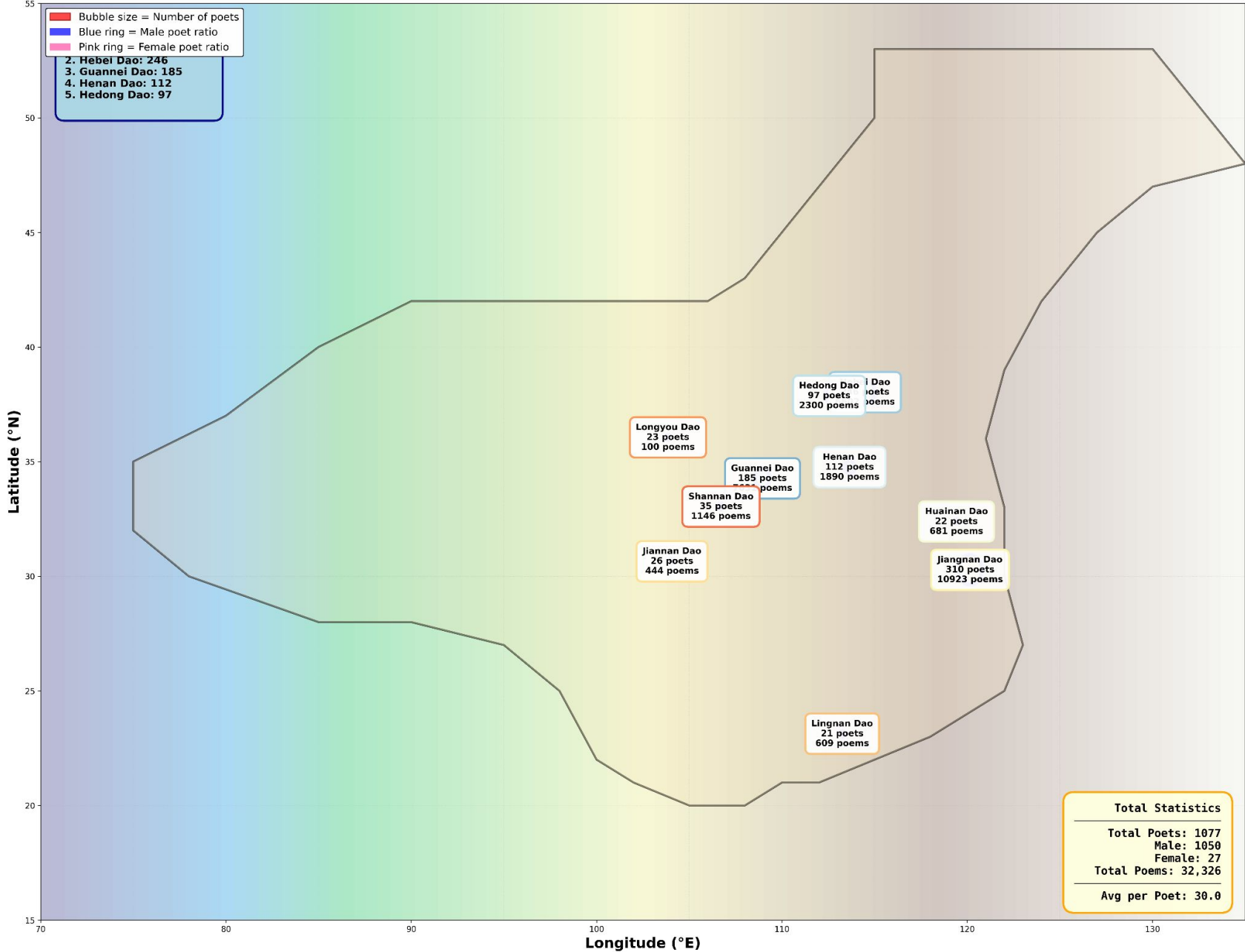
劍南 Jiannan

嶺南 Lingnan

♦ capital circuits (Chang'an 長安, Luoyang 洛陽). **Counts are heavily imbalanced — Jiangnan 126 poets, Longyou 6.**



Geographical Distribution of Tang Poetry Poets
By Tang Dynasty Ten Circuits (Dao)





Three tasks, three grains of geography



10 circuits

the full administrative map

hardest — many small classes



3 macro-regions

coarse regional grouping

intermediate grain



South 南 vs North 北

242 poets — the headline task

the axis literary history cares about

Imbalanced classes throughout → **balanced class weights, macro-F1 reported alongside accuracy.**

Style as countable character habits

Character n-gram TF-IDF. Count every character and every adjacent pair a poet uses; weight up what is distinctive to that poet's corpus, down what all Tang verse shares.

1–2 grams

 $\leq 8,000$ features

sublinear TF

Why it suits classical Chinese: no word segmentation, no parsing, no modern linguistic assumptions imposed on 8th-century verse — characters are the native unit.

A five-character line becomes ...

空 山 新 雨 後

“Empty mountain, after new rain” (Wang Wei)

UNIGRAMS

空

山

新

雨

後

BIGRAMS

空山

山新

新雨

雨後

Each chip is a measurable habit of style — TF-IDF asks which habits set one poet, or one region, apart.

Features a literary historian can read



Imagery classes

mountain 山 · water 水 · plant · fauna ·
celestial — relative frequencies per poet



Seasonal markers

temporal and seasonal references across the
corpus



Allusion density

canonical citations and references to
historical figures



Lexical richness

type–token ratio: how varied a poet's
character inventory is



Tonal patterns 平仄

approximated only — a complete
Middle-Chinese rime dictionary is lacking

Interpretability is the point — **each feature maps to a claim a Tang scholar could contest with close reading.**

Five readers, one fair test

Logistic regression

linear weights on features — fully inspectable

Linear SVM

maximum-margin linear separator

Random forest

ensembles of decision rules

MLP

a small neural network

GuwenBERT 古文BERT

a transformer pre-trained on classical Chinese



The fair test

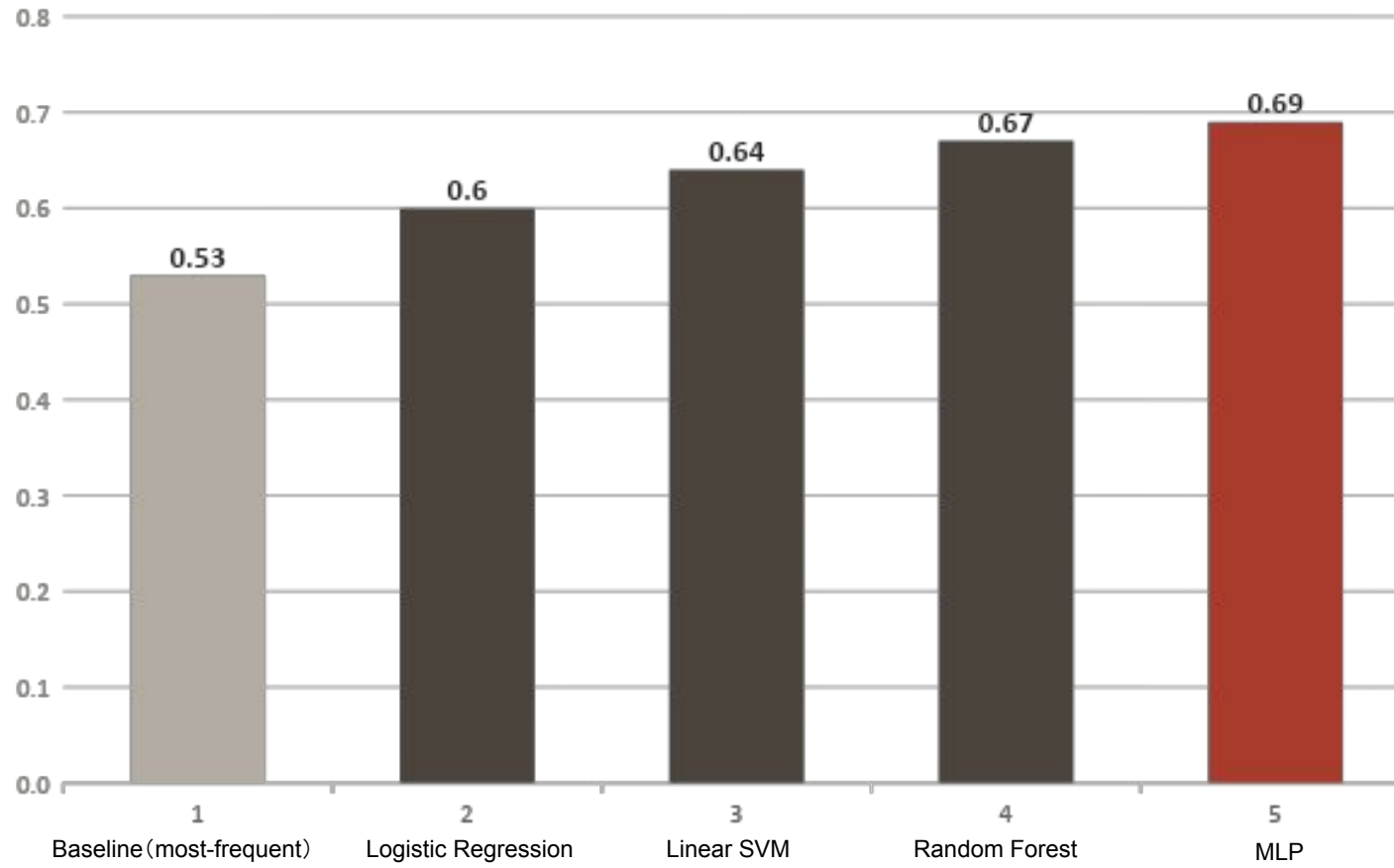
Stratified 5-fold cross-validation — every poet is a held-out test case exactly once

Balanced class weights — small circuits are not drowned out by Jiangnan

Accuracy + macro-F1 — macro-F1 treats every region equally, large or small

Baseline: most-frequent class — “always guess the biggest region” — 0.53 on South/North

Geography is legible from verse alone



South vs North classification · stratified 5-fold CV · 242 poets

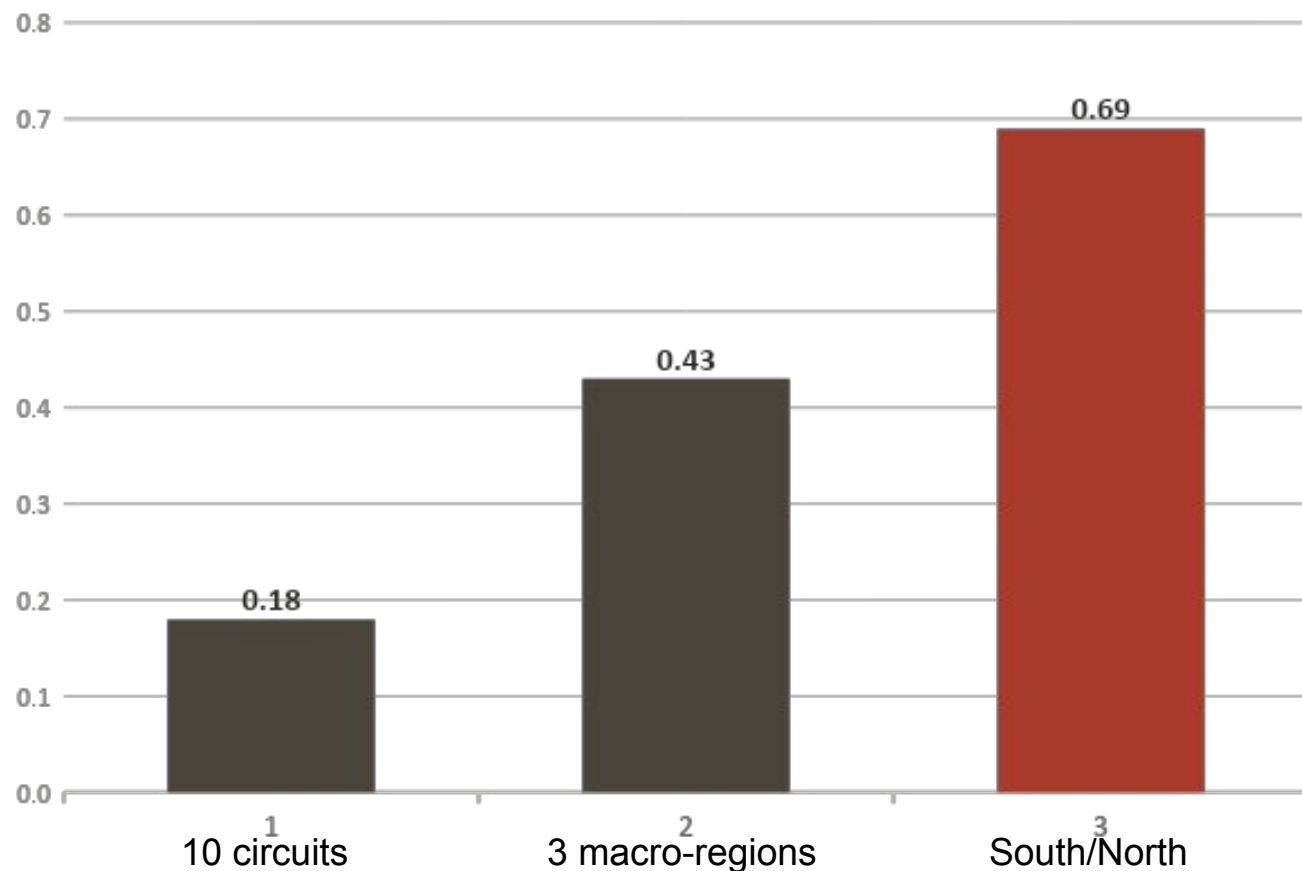
0.69

accuracy on South vs North — against a 0.53 most-frequent baseline

Macro-F1 0.69 vs 0.35 — the gain is not an artefact of class imbalance.

A poet's origin is readable, well above chance, from text alone.

The signal survives at every grain



Macro-F1 by task granularity · each $\approx 2\times$ its most-frequent baseline

Coarse maps are sharper — but even the hardest map is readable

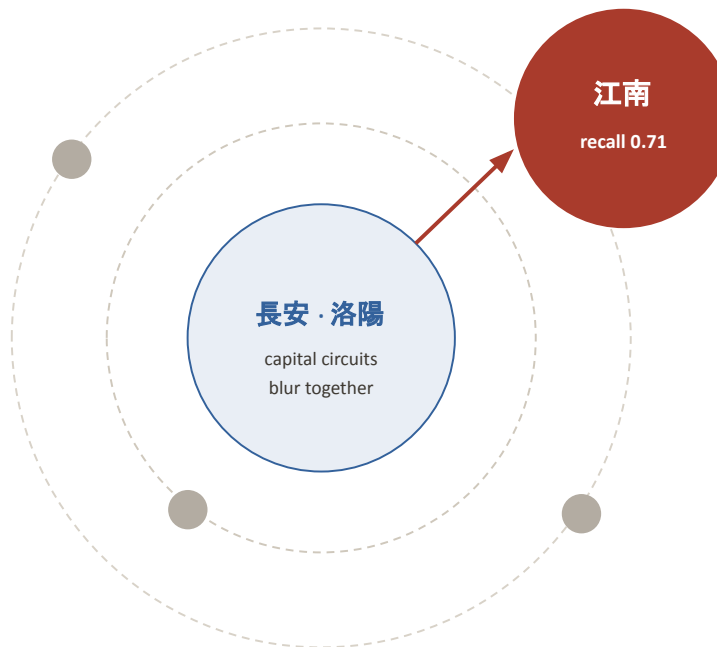
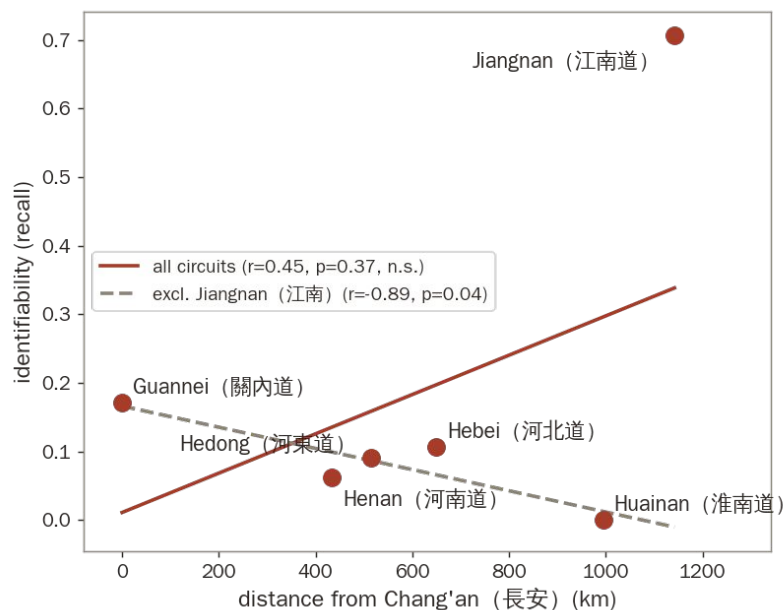
0.69 South / North — the axis of the historical debate

0.43 3 macro-regions — regional structure persists

0.18 10 circuits — $\approx 2\times$ baseline despite tiny classes

Ten-way prediction with six-poet classes is a brutal task — doubling baseline there is the surprise.

The centre blurs; Jiangnan stands apart



Near the court, one idiom. Circuits around Chang'an and Luoyang are heavily confused with one another — the shared court language erases local difference.

Far from it, a voice of one's own. Jiangnan — the empire's literary south — is by far the most separable region, at recall 0.71 on the ten-way task.

Per-circuit identifiability (recall) vs. distance from Chang'an — the apparent rise is carried by Jiangnan alone.

Pattern: Not a distance law. Across all six circuits the fit is non-significant ($r = 0.45$, $p = 0.37$); drop Jiangnan and the slope reverses ($r = -0.89$). Two facts, not one gradient: the capitals share a single court idiom — and Jiangnan alone keeps a voice of its own.

Imagery carries the region



The South writes landscape

Mountain & water imagery 山水 — used markedly more than in the north

Buddhist & recluse vocabulary — temples, retreat, withdrawal

Landscape diction — the discriminative characters of the south



The North writes the court

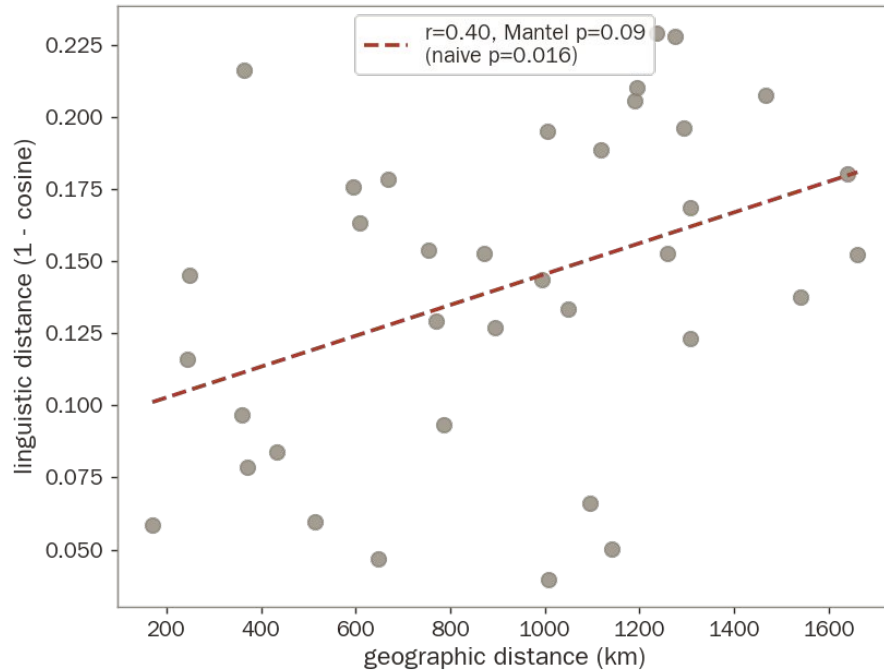
Palace diction — the built world of the capitals

Gongti 宮體 motifs — courtly “boudoir” poetry inherited from the southern dynasties' courts

Court-idiom characters — the discriminative characters of the north

Tonal patterns 平仄 carry comparatively weak geographic signal — **the regional fingerprint lives in what poets see, not in how their verses scan.**

Linguistic distance grows with the map



In plain terms: the farther apart two circuits lie on the map, the more distant their poetic language — dialect geography's classic pattern, surfacing in a literary corpus 1,300 years old.

Suggestive rather than decisive at this sample size.

SCHEMATIC



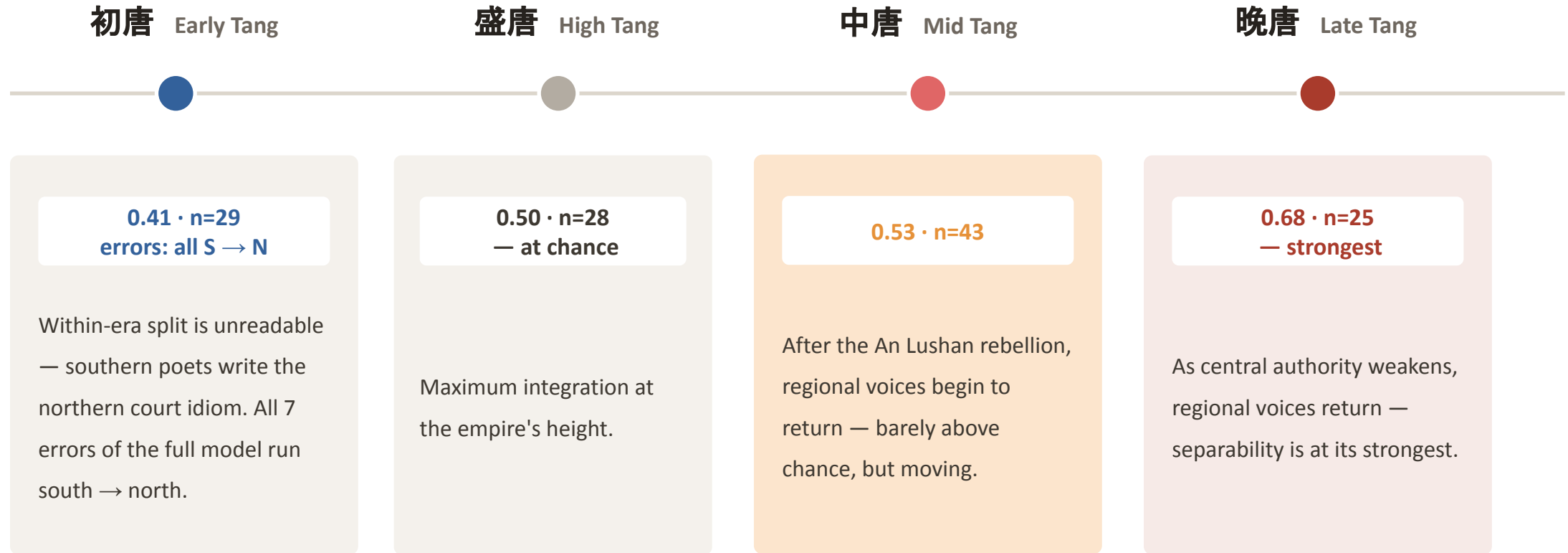
Neighbouring circuits → similar poetic language



Distant circuits → distinct poetic language

Distance measured between circuit centroids; linguistic distance between circuit-level stylistic profiles.

The fingerprint waxes and wanes with the dynasty



Regional distinctiveness is not constant — **it waxes and wanes with the political integration of the dynasty.**

When the model errs, it errs like a historian

7 / 7

Early-Tang misclassifications read a southern poet as northern — none err the other way.

虞世南

Yu Shinan

褚亮

Chu Liang

賀知章

He Zhizhang

Southern-born, court-trained — all three wrote in the dominant northern court idiom.

鄉音無改？

“My native accent is unchanged ...”

He Zhizhang swore the accent of home never left him. On the page, the classifier reads him as **northern**.

The “error” is the court's gravity, measured. Misclassification here is not noise — it encodes the prestige of the early-Tang court style.

Do transformers read anything more?

0.62 ± 0.06

Naive fine-tuning

GuwenBERT fine-tuned per fragment —
below the character n-gram models

0.674

Hierarchical frozen encoder

pool fragments over a poet's whole corpus
— now ties the best classical models
(0.67–0.69)

+ 0

Hybrid BERT + TF-IDF

combining the two adds nothing at all

Character n-grams already capture the available regional signal. For this corpus, interpretability costs nothing — the readable model and the deep model see the same geography.

What this offers literary history



A dynamic answer to the regional-schools debate

Regional distinctiveness is real but not constant — it waxes and wanes with political integration, now with numbers attached.



Hypotheses a scholar can contest

Because features are imagery, allusion, and diction — not opaque embeddings — every finding is a soft claim open to close reading.



Errors as evidence

The model's failures mark exactly where court prestige overrode origin — a productive human-machine division of labour.

Not a replacement for close reading — **an instrument that tells close readers where to look.**

Read with care



Small samples

357 poets overall, 242 for South/North — and era subsets are smaller still.



Inherited noise

Poem attribution and geographic labels carry the uncertainties of the Complete Tang Poems and CBDB.



Approximate phonology

Tonal 平仄 features are approximated — a complete Middle-Chinese rime dictionary is lacking.



Suggestive geography

Distance decay ($r = 0.40$, $p \approx 0.09$) needs prefecture-level geocoding and more poets to firm up.

All four point the same way

Finer geocoding, richer historical phonology, and more poets would sharpen — not overturn — the picture. The claims are sized to the evidence.

CONCLUSION

A Tang poet's origin leaves a computationally detectable trace in their verse.

- **Carried chiefly by imagery and lexical choice** — mountains and waters in the south; palace and gongti diction in the north
- **Decaying with geographic distance** — farther circuits, more distant language ($r = 0.40$)
- **Modulated by historical era** — at chance amid High-Tang integration; strongest (0.68) as Late-Tang authority wanes

Next: extending the temporal analysis, and stylistic similarity networks — does textual proximity track geography, or transcend it through literary influence?

謝謝 Thank you · arXiv:2606.24093 · Chi-Sheng Chen & Hung-Yun Liu

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